

storing and delivering mechanical power.

How does it work? Special synthetic compounds called elastomers, which have the stretching properties of natural rubber, are moulded into several patented shapes. Each shape



Astronaut Sunita L. Williams, Expedition 15 flight engineer, exercises on the Cycle Ergometer

creates a different type of resistance. By varying how these parts are used – alone, linked together, or used in combination with other devices – almost any type of exercise can be achieved.

- ▶ **Treadmill with Vibration Isolation System:** When astronauts on the space station use a treadmill (made specially to work in micro-gravity), spring-loaded cords attach to both sides of the treadmill and to a harness around the runner's waist. The cords can be loaded with 66 to 100 per cent of the runner's body weight to determine the strength of the workout.
- ▶ **Cycle Ergometer:** Sometimes astronauts on the space station exercise on a cycle ergometer – sort of a glorified stationary bicycle. The astronauts use clip pedals and have the option of waist straps, back supports, and hand holds to secure themselves to the machine and then pedal away with attached weight.

Moving air is supplied to dry off the perspiration produced from exercising. Otherwise, the sweat would stick to the skin and grow thicker and thicker.

Sleeping in Space

Just like on Earth, a worker in space goes to bed at night then wakes up the next day and prepares for work all over again. There are a few differences, though.

For starts, in space, there is no up or down and there is no gravity. As a result, astronauts are weightless and can sleep in any orientation. However, they have to attach themselves to a wall, a seat or a bunk bed inside the crew cabin so they don't float around and bump into something.

Crews usually sleep in sleeping bags. On the space shuttle, astronauts can also sleep in the commander's seat, the pilot's seat or in bunk beds. There are only four bunk beds in the space shuttle. So that means on missions with five or